

Recording a route without satellites

How accurate a phone’s own sensors are, measured against GPS

Yu Yao-Hsing
euler.yu@gmail.com

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Abstract

When GPS is lost — in a tunnel, an underground car park, or an aircraft — a phone can still draw the route it travels. It reads *speed* from how the vehicle shakes and *direction* from its gyroscope and compass, and it never uses GPS after the first point. This paper measures how well the current version of that system works, by recording GPS at the same time and using it only as the answer key. Over nine motorcycle journeys with the phone in a trouser pocket (2.8 hours, 84 km), the distance the system measured was within 0.4% of GPS in total, and within 16% on every journey. Whole routes were turned from their true direction by a median of 9°. The main weaknesses are speed at the extremes — fast riding reads slow and slow riding reads fast — and the first minute of a ride, before the phone has learned how it sits in the pocket.

1 What the system does

A route is a series of short steps. Each step needs a length and a direction. With GPS both come from satellites; here both come from the phone’s motion sensors.

Speed from vibration. A moving vehicle shakes, and it shakes differently at different speeds [1]. (Adding up acceleration instead does not work on a phone: its small sensor errors grow into kilometres within minutes [2].) Every second, the phone takes the last four seconds of up-and-down acceleration, measured 50 times a second, and turns it into a short “fingerprint” of eleven numbers — how much energy the shaking has at each of nine frequency ranges, plus two measures of its overall strength. Whenever GPS is available, the phone stores fingerprints together with the speed GPS measured at that moment; it keeps up to 4000 of them. Without GPS, it looks up the 12 stored fingerprints closest to the current one and averages their speeds:

$$\hat{v} = \frac{\sum_{j=1}^{12} v_j / d_j^2}{\sum_{j=1}^{12} 1 / d_j^2} \quad (1)$$

where v_j is a stored speed and d_j how far its fingerprint is from the current one. If even the closest stored fingerprint is far away, the phone does not guess. Averaging tends to pull every answer toward the middle, so a correction learned from the stored examples stretches the answers back out.

Direction from turns. The gyroscope measures every turn precisely but does not know north; the compass knows north but measures where the *phone* points, not where the vehicle is going — a known problem for phones carried in different ways [3]. In a pocket those differ by some angle β , which depends on how the phone happens to sit. The phone learns β from the vehicle’s own turns. In a turn the vehicle is pushed toward the inside of the curve, so the sideways push the phone feels, compared with which way the gyroscope says it is turning, shows where “forward” is. Written as complex numbers,

$$\hat{\beta} = \arg \sum_t \frac{a_t}{|a_t|} \overline{i v_t \omega_t} \quad (2)$$

where a_t is the sideways and forward acceleration measured relative to the compass, v_t the speed and ω_t the turn rate. The phone makes a first estimate after 30 seconds of riding and keeps improving it; when it improves, the part of the ride already drawn is redrawn with the better value.

Walking is handled differently: over four seconds, the back-and-forth of the steps lines up with the direction of walking, and each push-off is sharper than each landing, which says which end of that line is forward.

The compass reading itself. The phone’s built-in heading is unreliable in one common pocket position: when the phone sits head-down at an angle, it scattered by about 49° a minute even though the phone was not moving in the pocket. So the system computes its own heading from the phone’s orientation, using whichever edge of the phone is closest to level. That heading scattered by about 1° a minute in the same position.

Only the first point uses GPS. After that, each step is laid down from the previous point, along the current direction, for the distance the speed covers in that second.

2 How it was tested

GPS was recorded alongside every test and never used by the system; it serves only as the answer key. Two sets of recordings were used:

- **Nine journeys with the current version** (22–24 September 2026): motorcycle, phone in a trouser pocket, most ending with a short walk; 2.8 hours and 84 km of GPS track. These are used for speed, distance and whole routes, because they were recorded by exactly the system described above.
- **All 61 recordings with the phone’s orientation logged** (15 August to 24 September; car and motorcycle). The current direction method was run again over their sensor recordings, so these measure direction only. Seven include a walk long and straight enough to check.

In a pocket GPS is itself only accurate to 10–40 m, so it was used over stretches long enough for that not to matter:

- **Speed and distance** are compared over half-minute stretches between GPS fixes that are better than 40 m: the straight distance GPS moved, against the distance the system added.
- **Direction** is compared every half-minute: the bearing from the GPS position 15 s earlier to the one 15 s later, against the system’s heading over the same half-minute.
- **Whole routes** are compared by finding the rotation and the size change that best lay the drawn route over the GPS track, pinned at the shared starting point. The rotation is how far the whole route is turned; the size change is how much too long or short it is.

3 Results

3.1 Speed

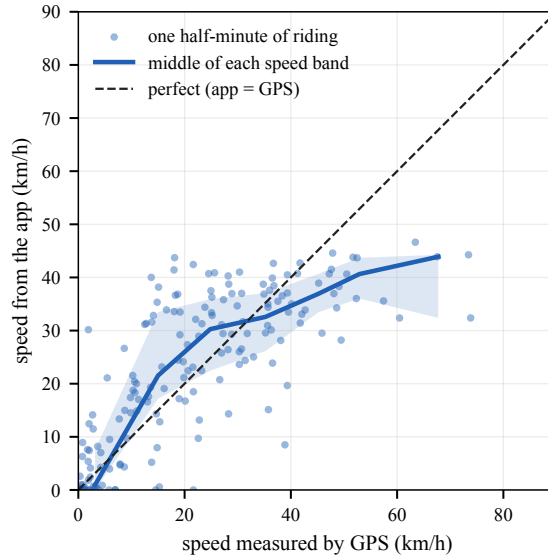


Figure 1: Speed from the system against speed from GPS, over 184 half-minutes of riding in the nine journeys. On the dashed line the two agree. The solid line is the middle of each 10 km/h band, and the shading holds the middle half of the readings.

Figure 1 shows the speed. Standing still and crawling are recognised correctly. In between, the system squeezes the range: 10–20 km/h reads about 8 km/h high, 30–40 km/h is close, and above 60 km/h it read about 42 where GPS measured 79. The average error is 8.5 km/h. The squeeze comes from averaging twelve examples: when a fingerprint could belong to several speeds, the average lands in the middle.

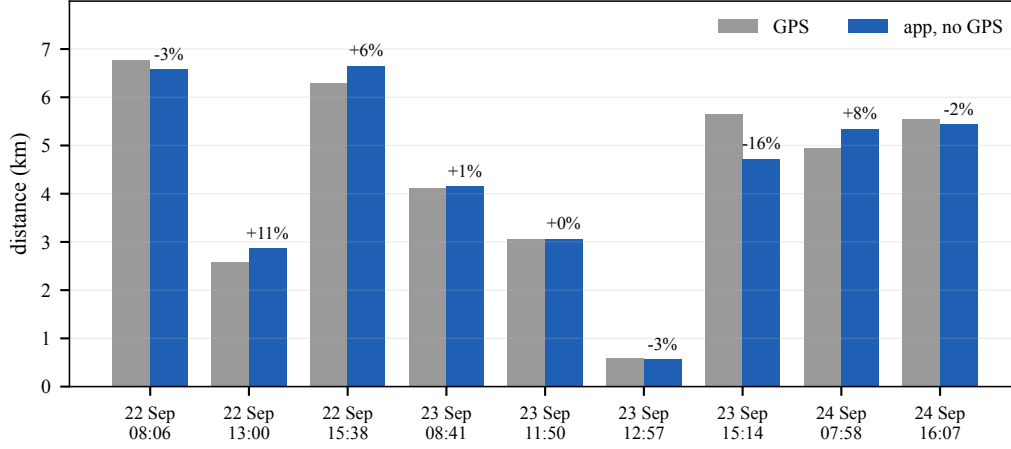


Figure 2: Distance over the stretches where GPS was good enough to check, for each journey. The percentage is the system’s difference from GPS.

3.2 Distance

The over-reading at low speed and under-reading at high speed largely cancel over a journey. Over the stretches GPS could check — 25–83% of each journey’s time — the system added 39.4 km where GPS measured 39.5 km, 0.4% short. Individual journeys ranged from 16% short to 11% long (Figure 2).

3.3 Direction

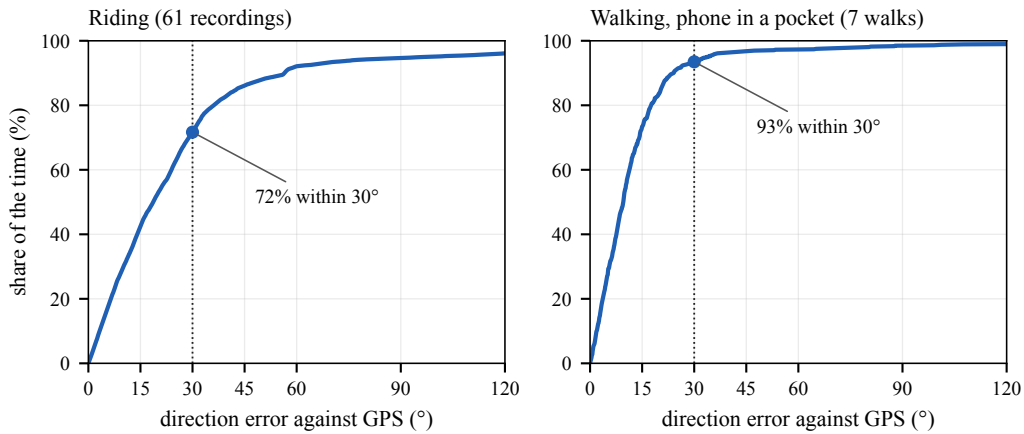


Figure 3: How often the system’s direction is within a given angle of the GPS direction. Left: 19,570 half-minutes of riding over 61 recordings. Right: 721 seconds of walking over seven walks with the phone in a pocket.

While riding, the direction was within 30° of GPS 72% of the time and within 45° 86% of the time; the middle error was 19° (Figure 3). While walking with the phone in a pocket it

was within 30° 93% of the time, with a middle error of 10° . Checked against the angle GPS implies, the pocket angle β learned from turns was typically within about 10° , and about 11° after only 30 seconds.

3.4 Whole routes

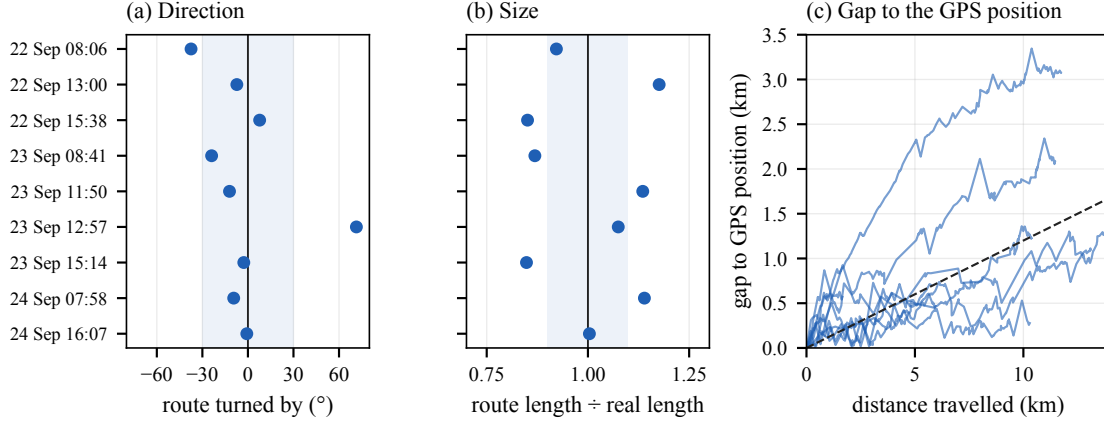


Figure 4: The nine journeys drawn by the system, compared with GPS. (a) How far the whole route is turned from the GPS track; the shaded band is $\pm 30^\circ$. (b) Its length relative to the GPS track; the band is $\pm 10\%$. (c) How far the drawn position is from the GPS position as the journey goes on; the dashed line is 12% of the distance travelled.

Table 1: The nine journeys. *Turned* and *Length* as in Figure 4(a,b); *Gap* is the distance from the GPS position, as a share of the distance travelled so far (the middle value over the journey).

Journey	Duration	GPS track	Turned	Length	Gap
22 Sep 08:06	32 min	11.7 km	-38°	0.92	46%
22 Sep 13:00	12 min	6.0 km	-7°	1.18	8%
22 Sep 15:38	19 min	13.9 km	$+8^\circ$	0.85	9%
23 Sep 08:41	21 min	11.5 km	-24°	0.87	19%
23 Sep 11:50	8 min	5.7 km	-12°	1.14	12%
23 Sep 12:57	8 min	1.7 km	$+71^\circ$	1.07	49%
23 Sep 15:14	27 min	13.1 km	-3°	0.85	12%
24 Sep 07:58	20 min	10.4 km	-9°	1.14	12%
24 Sep 16:07	20 min	10.3 km	-1°	1.00	6%

Figure 4 and Table 1 put speed and direction together. Six of the nine routes are turned less than 15° from the truth, and seven less than 30° ; the middle value is 9° . Lengths run from 0.85 to 1.18 of the real length, with a middle value of 1.00. The drawn position drifts away from GPS as the journey goes on — typically by about 12% of the distance travelled, so about 1.2 km after 10 km.

Two journeys stand out. The 8-minute journey at 12:57 contained under a minute of riding at speed, too little to learn the pocket angle well, and its route is turned 71° . The journey

at 08:06 on 22 September is turned 38° ; its pocket angle was learned about 25° off and stayed there.

4 Where it goes wrong

- **Fast riding reads slow, slow riding reads fast.** Above 60 km/h the speed can be half the true value (Figure 1). Over a whole journey the two partly cancel, but a journey that is mostly fast comes out short.
- **The first minute of a ride.** Until the pocket angle is learned, the direction comes from the compass alone, which in a pocket can be far off. The redrawing fixes the saved route afterwards, but a very short ride may never learn the angle well.
- **A phone that moves.** The pocket angle is assumed to stay the same for the whole ride. If the phone shifts, the rest of the ride is turned by about as much as it moved. When the phone is taken out of the pocket, the hand's movement hides the vehicle's vibration, so the last speed is held; on one ride, 48 km/h was held for ten seconds after the motorcycle had stopped, about 130 m that was never travelled.
- **Car-park ramps.** A car crawling up a concrete ramp shakes like one driving at 40 km/h, so distance is not counted while climbing and turning together. The same test can fire on an ordinary rising, curving road: once for nine seconds at 48 km/h, about 115 m lost.
- **Aircraft.** Speed cannot be measured in an aircraft. The phone's own orientation filter treats the long, steady push of a takeoff as a tilt [4], and the signal is gone.

5 What did not work

Many ideas were built and tested against GPS before the current version. The most instructive:

- **Neural networks.** A network that reads 20–40 seconds of fingerprints at once was more accurate on test data (8.8 against 10.6 km/h) and ran in the app for a while, but after it was retrained it under-read a pocketed ride by a third, and it was removed. Another network, better than the current system on familiar data, was confidently wrong on a flight it had never seen, recording 5% of the distance, where the current system simply declines to answer.
- **Direction from speeding up and braking.** The push when accelerating in a straight line points forward, so it should give direction. It was 44° off; turns work far better.
- **Other direction ideas:** the axis the motorcycle leans about (25° off), the direction of the rider's thigh (73° off), and the swing of the thigh for telling forward from backward when walking (wrong more often than right).

- **How examples were forgotten.** When the store was full, it used to throw away examples from whichever speed it held most of, to keep rare fast examples. That left it holding 5% of its examples below 5 km/h when 24% of real riding is that slow, so slow riding read far too fast. Throwing away a random example instead fixed most of it.

6 Lessons

- **The phone’s own definitions caused more error than the models.** Its built-in heading becomes unreliable in a common pocket position, and its “user acceleration” points the opposite way to the vehicle’s acceleration. Each of these, once found, mattered more than any change to the estimators.
- **A test must not use what it is testing without.** Twice during development the no-GPS test mode quietly used GPS for direction, which made its results look better than they were.
- **Check what a learning system remembers.** The store’s way of forgetting examples was sensible on its own and still shifted every answer, and it went unnoticed until its contents were compared with the riding they came from.

Limits of this evaluation

All recordings were made by one person, on one iPhone, in one city. The speed, distance and whole-route results come from nine motorcycle journeys with the phone in a pocket; cars and other carry positions may behave differently. GPS in a pocket is itself only accurate to tens of metres, so small errors cannot be measured reliably. The direction results for older recordings come from running the current method again over their sensor logs.

References

- [1] A. Cohen and I. Klein. CarSpeedNet: Learning-based speed estimation from accelerometer-only inertial sensing. arXiv:2401.07468, 2024.
- [2] Smartphone MEMS accelerometer and gyroscope measurement errors: laboratory testing and analysis of the effects on positioning performance. *Sensors*, 23(17):7609, 2023.
- [3] Pedestrian dead reckoning with novel heading estimation under magnetic interference and multiple smartphone postures. *Measurement*, 183, 2021.
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